

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

Graph Neural Networks for Constructing (k, g) -
Graphs
Bachelor Thesis

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This section will be completed near the end of the thesis.

1 Abstract

This thesis investigates the use of graph neural networks as learned heuristics for constructing small regular graphs of prescribed girth, also known as (k, g) -graphs. The final abstract will summarize the problem, proposed method, experiments, and main findings once the experimental contribution is fixed.

Keywords: graph neural networks, algebraic graph theory, cage problem, (k, g) -graphs, voltage graph lifts, heuristic search

Contents

1 Abstract	ii
2 Draft Scope	1
Bibliography	2

List of Figures

List of Tables

2 Draft Scope

The thesis concerns the construction of (k, g) -graphs: k -regular graphs whose shortest cycle has length g . The classical cage problem asks for such graphs with the smallest possible order. Known approaches include algebraic constructions, voltage graph lifts, exhaustive and heuristic search, and excision techniques [1], [2].

The current working direction is to use graph neural networks not as unrestricted graph generators, but as learned components inside constrained search procedures. This matches the structure of the problem: regularity and girth can often be enforced, checked, or pruned directly, while the learned model can score partial states, voltage assignments, or local graph edits.

The exact chapter structure is intentionally left open until the experimental contribution is stable.

Bibliography

- [1] G. Exoo and R. Jajcay, “Dynamic Cage Survey,” *The Electronic Journal of Combinatorics*, 2013.
- [2] G. Exoo, J. Goedgebeur, J. Jooker, L. Stubbe, and T. Van den Eede, “New small regular graphs of given girth: the cage problem and beyond.” 2025.